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A Hybrid Process Mining Approach for Business Processes in Financial Organizations

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Abstract

Business process management is an integral part of many organizations in doing their day-to-day business operations considering that they help organizations transfer their business critical information from one organizational entity to another in the form of automated state transitions. The data (or event logs) that have been generated during these transitions hold valuable insights for improving organizational business processes, highlighting problem areas and visualizing the actual vs the formal procedures. The goal of this paper is to summarize the development of a hybrid analytical approach that utilizes both SQL and NO-SQL based back-end platforms in harmony in order to carry out process mining for a participation bank in Turkey. For this purpose, first we have developed a hybrid analytical software infrastructure that is backed by MS SQL Server and Hadoop platform components in order to discover key business processes of the organization based on event data. We then established a process mining framework that visualizes the process performance indicators and proposes workflow design changes and carries out statistical tests for identifying performance fluctuations by particularly using an in-memory parallel processing framework, named Apache Spark.

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1. Introduction

In recent years, the advances in the communication technology, combined with social media, has led to the advent of “Big Data Era” where large volumes and variety of data are produced at extremely large speeds. As a consequence, the Data science has gained attention as a combination of multiple disciplines, such as statistics, data mining, and distributed systems.

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Among these multiple disciplines, Data Mining is particularly studies the relationships in large data sets (i.e. so called Big Data) with the intention to uncover the insights that are hidden in the data. On the other hand, Process Mining has emerged as a sub-discipline of data mining and it aims at combining the data analysis with modelling, control and improvement of business processes.

In simple terms, a business process can be defined as a sequence of logical activities that are executed towards process-oriented object(s), such as invoice and purchase order, in a completely closed and timely manner in order to fulfil a particular business need of an organization. Examples to business processes can be the loan assignment process in a bank, the routing business of a retailer, and the order processing of a factory [1]. The correct execution of business process are central to the organizational success considering that they deal with critical business functions.

In modern organizations today, the execution of business processes are handled by large scale IT systems. These systems generate business related event data (logs) while they execute the individual actions that are encapsulated by the business processes. These event data can contain valuable insights for improving organizational business processes, highlighting problem areas and visualizing the actual vs the formal procedures while running the business. Hence, they can be utilized for process mining.

So mentioned event data are generally stored in relational database storage systems (RDBMS). RDBMSs are designed for processing transactional data that is generated at a much limited volume and velocity. On the other hand, the insights that can be acquired from event data have “Big Data” characteristics, such as high volume (e.g. in the magnitude of multi-petabytes) and high-speed processing capability. Therefore, there exists a need to utilize both relational database systems (e.g. MS SQL Server) and non-relational NO-SQL systems (e.g. Hadoop Platform [20]) in order to acquire insights from the event data.

From above outlined perspective, in the study that is outlined in this paper, we have developed a hybrid analytical approach that utilizes both SQL and NO-SQL based back-end platforms in harmony in order to carry out a practical process mining work for a participation bank as a case study. For this purpose, we have developed a hybrid analytical software infrastructure that is backed by both MS SQL Server and Hadoop platform components [20] in order to discover key business processes of the bank based on event data.

The rest of this paper has been divided in 4 sections. In the section 2, the related research in the area of event data based process-mining has been described. In section 3, our case study has been described together with our results. In section 4, our main conclusions have been provided.

2. Related Work

In the process mining literature, a number of studies have been made about the development of techniques and algorithms that make use of event data (i.e. event logs). One of the very first process mining studies that utilize event logs was described, for example, in [2]. In this study, the authors developed process mining techniques that are particularly based on control flow and case data in order to analyse the invoice handling of a local public services office.

In another event data based process mining study at [3], the authors have proposed a genetic process mining strategy for mitigating the noise in the event logs for generating realistic process models from event data that was obtained from Dutch municipalities. The authors claim that the genetic approach they developed made it possible to realize the mining of process models that are complete (can reproduce all the behaviour in the log) and precise (do not allow for extra behaviour that cannot be derived from the event log) [3].

Bozkaya et al. [4], proposed a six-phase process diagnostics methodology and validated it in a case study in public services sector. In addition, Weerdt et al. [5] developed practical approach to even data based process mining through a case study in financial services sector. This was one of the very first structural process mining methodologies for dealing with the real-life event logs that exhibited much less structured process behaviour. The technique that was proposed in this methodology made use of different perspectives of event data, namely, control-flow, organizational and case data perspectives. Compared to the six-phase methodology that was proposed in [4], the methodology in this paper [5] particularly focuses on data extraction and exploration as well as on the analysis of process execution data in financial services. In our case study that is presented in the next section we particularly benefited from the proposed methodology at [5].

Another event data based process mining study that is particularly in health-care sector was presented by Rebugue and Ferreira in [6]. In this study, the authors developed a methodology that leads to the discovery of medical processes involving regular behaviour, process variants and exceptional cases at a hospital emergency service. This methodology integrated the main stages of process analysis specifically in health-care.

Aalst et al. [7], on the other hand, conducted a process mining study again using real-life event logs in order to estimate the time for a business process to complete. The authors demonstrated that the process models that were extracted from event logs can be extended with information based on the predicted completion time of the associated business processes. Weerdt et al. [8] presented a multi-dimensional quality assessment technique in order to evaluate the process discovery techniques that make use of event logs extracted from information systems. The results of the study that is described in this paper shows that the *HeuristicsMiner* algorithm is particularly suited in real-life settings. However, the results also show that knowledge discovery from complex event log records can become a major problem for traditional process discovery techniques.

There are also studies that involved the use of distributed parallel processing frameworks like *Hadoop MapReduce* [21] in order to do process mining based on event logs. Meddah et al. [9],[10], for example, applied *Hadoop MapReduce* framework to process mining in order to extract information from the process traces that were obtained from the execution of two software applications, namely, Skype and Viber. In this work, the authors first developed a technique that discovers small patterns from the execution traces of these applications by using regular expressions. Secondly, they computed the discovered patterns and combined them based on composition rules by using parallel processing capabilities of *Hadoop MapReduce* framework. In contrast to the work that was described in these papers at [9] and [10], we have specifically made use of a fast in-memory distributed parallel processing framework namely *Apache Spark* [11] in our case study that is presented in the next section.

3. Case Study

3.1. Motivation for the Hybrid Approach

Banks today are in the middle of many financial operations considering that they enable organizations to exchange assets and/or services as part of their business processes. Banks utilize large scale distributed information systems in order to conduct these financial operations and these systems can produce large amount of transactional data as a result of the financial operations. The individual steps that are involved in a transaction (i.e. an event such as withdrawing a certain amount of money from a saving account) generate what we call as the “event logs”. These logs can be used as input to perform analytics study that can let us discover insights for improving organizational business processes, highlighting problem areas and visualizing the actual vs the formal procedures.

Due to the transactional nature of the operations that banks perform, the so mentioned event logs are generally stored in relational database management systems (RDBMS) like MS SQL Server. This is because RDBMSs provide guaranteed execution of transactions. This guaranteed execution means the individual steps that are involved in a transaction (i.e. events) are executed persistently either as a whole or not at all as a consequence of the ACID properties that are present in RDBMSs (i.e. Atomicity, Consistency, Isolation and Durability) [12].

Although the RDBMSs are specialized in managing transactional data, they have certain limits in processing large amount of data (i.e. Big Data) that have certain characteristics in terms of volume, velocity and variety. The term volume here refers to the size of the data in ranges from terabytes to petabytes [13]. Velocity, on the other hand, can be conceived as the speed of data capturing, processing and visualizing [13]. Variety refers to the types of data such as structured (e.g. web log data), un-structured (e.g. Facebook comments), and semi-structured (e.g. JSON, XML data) [13].

In contrast to the relational database systems, the non-relational platforms (e.g. Hadoop [20]) provide much better tools and frameworks, such as *Hadoop MapReduce* [21], *Apache Hive* [23] and *Apache Spark* [11] that are specifically designed for the storage and parallel processing of large variety and volume of data that can be in structured, unstructured and semi-structured format (i.e. big data).

Considering the transactional nature of the event logs in the banking sector, which are produced by the underlying information systems, RDBMSs are predominantly in use for storing such event logs. In order to fetch these event logs and turn them into knowledge for improving organizational business processes, one needs to make a combined use of

both RDBMSs and non-relational platforms (e.g. Hadoop [20]) in harmony. From this perspective, in the work that has been presented in this paper, we have developed a hybrid process mining approach that utilizes both SQL and NO-SQL based back-end platforms in harmony in order to carry out a practical process mining work for a participation bank as a case study.

We have divided our process mining study over a number of stages namely “*Data Preparation, Data Grouping, Data Filtering, and Variance and Regression Analysis*”.

3.2. Context

The participation bank where we have conducted our case study makes use of a distributed software system, named BOA Banking Platform [14]. This platform has been built in-house to meet the technology and business requirements of an international multi-channel bank. Examples to so mentioned multi-channels are mobile banking, Internet banking, branch banking (i.e. traditional banking) and API banking [15].

The BOA platform includes an in-house developed work-flow engine executing the work-flow instructions that are used to perform the multi-channel banking operations. It provides a graphical user interface (GUI) where business people (e.g. operation managers - OPMs) can visually draw a workflow with the support of developers as shown in figure 1 below. In essence, this drawing in figure 1 constitutes a blue-print that defines how a specific business operation can be executed. In other words, the drawing represents a state-machine diagram that instructs the workflow engine about what actions can be taken at each state under certain business related circumstances.

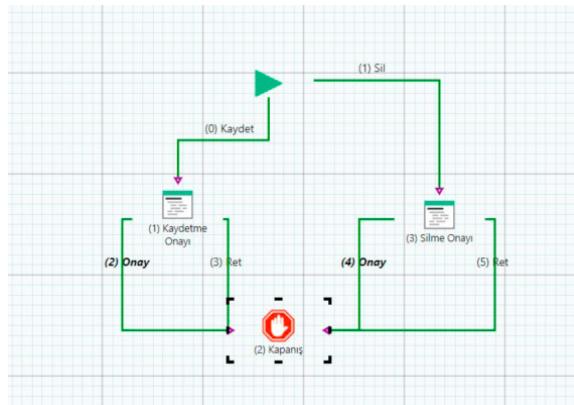


Fig. 1. Sample Workflow Design in BOA Workflow Engine

Hence, we have based our process mining study on the *event logs* that have been automatically generated by the above mentioned *work-flow engine*. This engine stores all relevant event log data in semi-structured format in *relational tables* in *MS SQL Server*. The event log data include what actions (i.e. events) are taken out of a particular state as part of a specific work-flow description that implements a banking operation, such as “*Financial Guarantee Letter Issuance*”. These actions and the states from where the actions are emerging have been illustrated by the arrows and squares, respectively, in figure 1 above. For example, the state “*Kaydetme Onayı*” (i.e. literally “*Approval of Persistence*”) is represented by a square in the lower left area in figure 2. On the other hand, the actions “*Onay*” (i.e. *Approval*) and “*Ret*” (i.e. *Reject*) are represented by the arrows that are emerged from the state “*Kaydetme Onayı*”.

3.3. Data Preparation

In this very first stage of our process mining study, we firstly investigated the so mentioned relational tables that hold the raw event log data in MS SQL Server. Upon our initial investigation, we created yet another table (so called base table) in MS SQL Server and gathered all relevant event data into this table whose schema definition is provided in table 1 below. The gathering of data has been done through a stored procedure that traverse through all the related lookup tables containing event logs

Table 1. Schema Definition (i.e. structure) of the base relational table that holds the event logs in structure on which we base our process mining study

COLUMN NAME	TYPE	DESCRIPTION
InstanceId	int	The specific instance of the workflow
InstanceStateId	int	The specific instance of the state to which an action has been taken
FlowId	int	Holds the identifier of the specific workflow design
StateId	int	Holds the identifier of the state to which a transition occurred
UserCode	varchar	Identifier of the system user (i.e. human being) who triggered the state transition
ActionId	smallint	Identifier of the action that leads to the state transition
InboxDuration	bigint	Total waiting time in the system user's inbox
WorkingDuration	bigint	Total time the system user has performed work in that specific state in milliseconds
StateStartDate	datetime	The date at which the state has been entered through a respective action
StateEndDate	datetime	The date at which the state has been leaved
PoolDuration	bigint	Total waiting time in the work of pool system user's work group (i.e. department)
StateDuration	bigint	Sum of InboxDuration + WorkingDuration + Pool Duration
InstanceRunId	int	Identifier of the specific workflow instance that is being executed

Secondly, we made use of an open source tool, named *Apache Sqoop* [22], in order to copy all the gathered event data in our base table in SQL Server into a Hive table that is present in our Hadoop cluster as illustrated in the figure 2. *Apache Hive* [23] is an open source NO-SQL database that works on top of *Hadoop Distributed File System* (HDFS) [24] and presents a schematic abstraction (e.g. view) to the underlying HDFS files that keep the real copied data (i.e. event data).

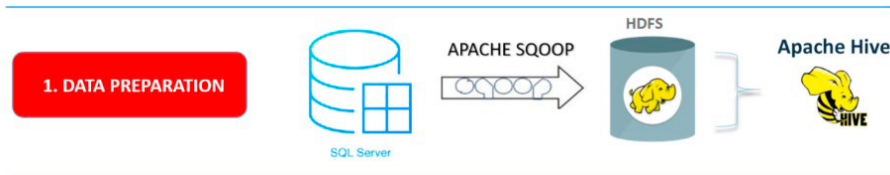


Fig. 2. Illustration of the batch copying of event data from SQL Server to Apache Hive

In order to instruct the *Apache Sqoop* about how the event data should be batch copied from our base table in SQL server into our *Hive* table in Hadoop, we issued below listed sqoop import command in our Hadoop cluster. As seen in the arguments of this command chain, the sqoop is instructed about the *URL of the SQL server*, the name of the *base table* and the other needed credentials in order to properly perform the batch copying procedure for the offline event data in to our Hadoop cluster for later offline batch processing.

```
sqoop import -Dhadoop.security.credential.provider.path=
jceks://hdfs/user/processmininguser.password.jceks
-D mapreduce.map.memory.mb=8196
--driver com.microsoft.sqlserver.jdbc.SQLServerDriver
--hive-import --hive-overwrite --hive-database processmining --hive-table BaseEventLogTable
--connection-manager org.apache.sqoop.manager.SQLServerManager
--connect 'jdbc:sqlserver://xyz.com;databaseName=XYZ'
--username pprocessmininguser
--password-alias processmininguseralias
--table BaseTable
--as-parquetfile
--target-dir /sqoopimports/processmining/ BaseEventLogTable
--compression-codec snappy -m 1 -
--table-hints NOLOCK --schema PM
```

3.4 Data Grouping

In this stage, we partitioned our event data, which was copied into *Hive* database in the previous stage, based on the fields *StateDuration*, *WorkingDuration* (See table 1) and the number of state instances, as illustrated in figure 3 below, in order to prepare data groups on which we performed variance and regression analysis that are described in

the following sub-sections.

2. DATA GROUPING									
flowId	stateId	usercode	NumOfStateInstances	flowId	stateId	usercode	statestartdate	stateenddate	duration
166	46	güzel	2083	166	95	evuksel	1362754437917	1362754607580	170
166	46	favdemir	1067	166	95	evuksel	1362754945903	1362755218430	10
166	46	vpabuccu	2345	166	95	ilknurc	1372315605063	1372315669857	64
166	46	oyanar	2567	166	95	ilknurc	1372341402297	1372341432610	30
166	46	hvaras	2098	166	95	evuksel	1363098179230	1363098260523	81

Fig. 3. Illustration of Data Grouping

Group 1 – StateDuration: The purpose of this data group (i.e. data set) was to calculate the total state duration, which is the sum of the values that are kept in the fields *PoolDuration*, *InboxDuration* and *WorkDuration*. For this purpose, we have grouped our data according to following setup. The descriptions for the specific fields in this setup are available in table 1.

```
+-----+-----+-----+-----+
|FlowId|StateId|UserCode|StateDuration|
+-----+-----+-----+-----+
```

Group 2 – WorkingDuration: The purpose of this data group was to calculate the working duration in which a system user spent time by working in a specific state of a specific workflow instance. Hence, we have grouped our event data by using the fields that are listed as follows:

```
+-----+-----+-----+-----+
|FlowId|StateId|UserCode|WorkingDuration|
+-----+-----+-----+-----+
```

Group 3 – NumOfStateInstances: The purpose of this data group was to calculate how many times each individual system user performed work in a specific state of a specific workflow instance. For this purpose, we have grouped our event data by using the fields that are listed as follows:

```
+-----+-----+-----+-----+
|flowId|StateId|usercode|NumOfStateInstances|
+-----+-----+-----+-----+
```

3.5 Data Filtering

In this stage, for each of the group that we have identified in the previous subsection, we detected the outlier data points and then left them out from the datasets in the respective groups by applying below listed step-wise procedure [16] for each pivot field namely, *StateDuration*, *WorkingDuration* and *NumOfStateInstances* as illustrated in figure 4 below. We have implemented this procedure by using a state-of-the-art big data processing framework for structured data, named *Apache Spark* [11], in the Java programming language. We have published this Java code implementing the procedure under MIT license at [17].

- i. First and third quantiles are found in the dataset based on the data points in the pivot field.
- ii. Inter-quantile range (IQR) is calculated by using the first and third quantiles.
- iii. Lower bounds and upper bound values are calculated based on the IQR, first quantile and third quantile as follows:

$$\text{lowerRange} = \text{firstQuantile} - 1.5 * \text{IQR}$$

$$\text{upperRange} = \text{thirdQuantile} - 1.5 * \text{IQR}$$
- iv. Leave out all of the data points that are NOT in between lower bound and upper bound (i.e. outliers).



Fig. 4. Illustration of the Procedure for Filtering out the Outlier Data Points

3.5 Variance and Regression Analysis

In this stage, we divided each group that were identified in sub-section 3.4 into further categories based on the field *UserCode* (see table 1) in order to analyze the groups in terms system user performance

Our intention here was to see how far the individual system users perform differently while they work on a specific state of each and every workflow instance that is executed by the work-flow engine. This was, in essence, for the purpose of identifying where bottlenecks occurred in the business operations and which system user contributed the most to the occurrence of these bottlenecks. To realize this intention, we made use of a statistical technique, named *One Way Analysis of Variance (ANOVA)* [18]. We implemented this technique again by using the *Apache Spark* framework [11] in the Java programming language and published the Java code again under MIT license in the GIT repository that is available at [17].

In addition to variance analysis, we also performed a regression analysis on our pivot fields namely, *StateDuration*, *WorkingDuration* and *NumOfStateInstances*, in order to see whether the values in these fields have a linear relationship with time. For this purpose, we have implemented a piece of code that performs linear regression by using the *Apache Spark* framework [11] in the Java programming language and published the code again under MIT license in the GIT repository that is available at [19].

3.6 Project Deliverables and Results

We have provided a brief summary of what have accomplished thus in this subsection although our study has not been fully finalized yet. At the SQL Server side, a tabular model report was generated under MS SQL Analysis Server and a Power BI report was prepared which showed us the basic process data – such as operation durations, occurrences etc. - of each the business flow (i.e. workflow). Although the BOA platform already provides semi-structured process data (or event log) that are in good-shape for analysis, we anyway created an MS SQL tabular model (i.e. the base table whose fields are described in table 1) in order to keep the necessary process measures calculated at hand. Then, we enhanced the basic process reports with more sophisticated graphs, such as box-plot, outlier detection and histogram.

It is very common that one process triggers another process one after another in financial business processes. For this reason, a big picture that visualizes the dependencies among the activities (i.e. stations) is also needed. We used a Power BI plug-in in order to visualize this big picture as a process report, which gave us a quick insight about what was going on in the end-to-end processes which is called the “Spaghetti View” in process mining terminology. The report provides the following measures:

- Average, min, max of Pool Duration, Inbox Duration and Working Duration
- Ratio of Pool Duration, Inbox Duration and Working Duration
- Average Working Duration per user

All above measures were dynamically calculated at *process level*, *station level* and the *activity level* for each state (or station). Example to the so mentioned “Spaghetti View” report has been provided at figure 5 as follows.

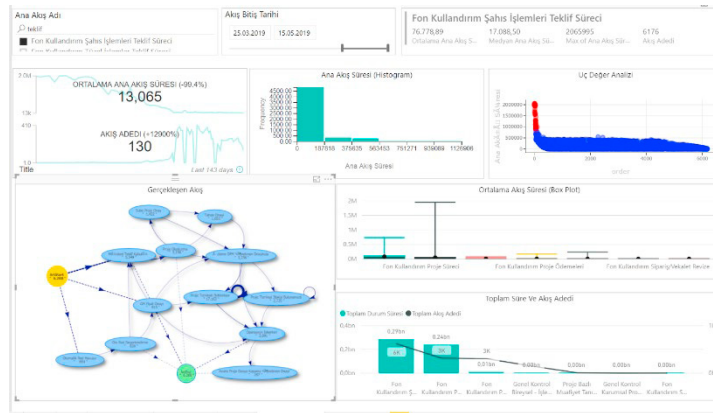


Fig. 5. Illustration of the Power BI Process Report Example – ‘Spagetti View’

While the Power BI report gave us detailed calculations of process measures, we also built another process view on top of our workflow module to present processes in simpler view. We calculated and showed the percentage of the instances that used a particular activity on the process design itself. This is so called ‘‘Lasagna View’’ and it was to help users to focus on a single process. We also developed process improvement suggestions to identify idle stations and activities, bottleneck stations and howed them in this lasagna view as shown in figure 6 below.

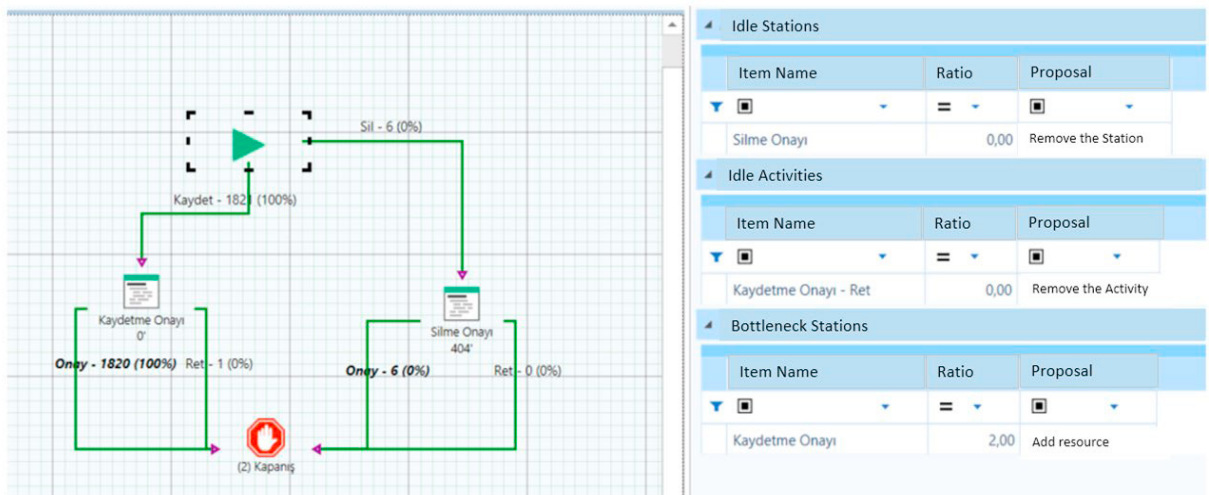


Fig. 6. Example Report for the Calculations – ‘Lasagna View’

At the Hadoop platform side, we tried to set-up an ‘‘alert mechanism’’ to flag the potential problematic processes. We used linear regression in order to capture the increase in station durations (i.e. time spent while in that station). On the other hand, we used ANOVA in order to see whether there were any differences in the average working durations of the users. We examined 67 process-station combinations that covers the duration of last 1 month and 3 months as the date intervals.

We have found that linear regression results are promising as summarized in table 2 and 3 below. Results show that most of the process station durations are stable. Only the durations for 12 stations showed positive trend based on one month data. Magnitude of the change was also crucial along with the significance. Table 3 below presents both the coefficient of the change and the p value. In this table, stations with p values that are higher than 0.1 and with very low coefficients are considered to be in green flag. Stations having statistically significant coefficients (p value < 0.05) and very low coefficients (coefficient less than 0.001) are considered to be in orange flag. These stations show very little increase in durations but the increase is statistically significant. Therefore, they need to be checked again. Stations

with high and significant coefficients are considered to be in red flag and must be checked first place. If there exists a high coefficient but no significance, then we consider it as yellow. Results show there are 2 stations in “red” condition and 17 stations in “orange” condition.

Table 2. Number of Stations based on Significance

Result	1 Month	3 Months
No Significant Duration Change	54	48
Duration increase at 0.05 significance	12	19

Table 3. Number of Stations by Alert Type

Coefficient	Significance	
	Not Significant	Significant
Low (Less then 0.001)	1 Month Data: 22	1 Month Data: 10
	3 Months Data:10	3 Months Data: 17
High	1 Month Data: 34	1 Month Data: 2
	3 Months Data: 21	3 Months Data: 2

On the other hand, ANOVA results that compare the average working durations of the system users turned out to be significant at almost all stations. Therefore, these results are not for practical use at the moment. Nevertheless, additional features might be needed, such as experience of the employee, education level etc. in order to achieve useful results.

4. Conclusions

Process mining is gaining attention day by day but is not a mature area regarding which methods will be utilized and how a proposal mechanism will be set. Much of the work is related to process discovery from event logs but there is not much how to detect specific problems at resource, product level etc.

During this study we developed a hybrid approach to address process mining activities that takes place at MS SQL side and at Apache Hive which is an open source NO-SQL database that works on top of Hadoop Distributed File System (HDFS).

We found that the use of Apache Spark was convenient for developing the analytical framework for statistical calculations. Nevertheless, we found that it would be probably better if we had prototyped the framework in a single PC first by using an easier to use library like numpy before attempting to develop things in the clustered environment using Apache Spark directly. Although spark is rather strong and state-of-the-art platform, its statistical libraries are not mature enough yet especially in the Scala and Java languages. Therefore, we spent some extra time to implement ANOVA like statistical calculations, by ourselves. We also tried to build an alert mechanism with linear regression showing which of process station durations tend to increase. This study is not finalized yet and we are still working to set-up a more sophisticated problem detection system.

At MS SQL side we composed visual reports via MS Analysis Server tabular model and Power BI. Plus we enhanced our internal workflow design module in a way to visualize the actual paths of process along with some proposal mechanism based on the frequency and durations of each station and activity.

References

- [1] Wil van der Aalst (2016) , “Process Mining, Data Science in Action, 2nd Edition”, *Springer Publishing*
- [2] Jochen De Weerd, Annalies Schuur, An Vanderloock, Bart Baesens (2012), “Process Mining for multi-facted analysis of business processes – A case study in financial services organization”, *Journal of Computer in Industry*, 64 (1) (2012), 57-67
- [3] A.K. Alves de Medeiros, A.J.M.M. Weijters, W.M.P. van der Aalst (2007), “Genetic process mining: an experimental evaluation, Data Mining and Knowledge Discovery”, *Journal of Data Mining and Knowledge Discovery*, 14 (2) (2007), 245–304

- [4] M. Bozkaya, J. Gabriels, J.M.E.M. van der Werf (2009), “Process diagnostics: a method based on process mining”, In A. Kusiak, S. Lee (Eds.), *eKNOW*, IEEE Computer Society, pp. 22–27
- [5] W. M. P. van der Aalst, H.A. Reijers, A.J.M.M. Weijters, B.F. van Dongen, A.K. Alves de Medeiros, M. Song, H.M.W. Verbeek (2007), “Business process mining: an industrial application”, *Journal of Information Systems*, 32 (5) (2007) 713–732.
- [6] Álvaro Rebugue, Diogo R. Ferreira (2012), “Business process analysis in healthcare environments: A methodology based on process mining”, *Journal of Information Systems*, 37 (2) (2012), 99–116
- [7] W. M. P. van der Aalst, M.H. Schonenberg, M. Song (2011), “Time prediction based on process mining”, *Journal of Information Systems*, 36 (2) (2011), 450–475
- [8] Jochen De Weerd, Manu De Backer, Jan Vanthienen, Bart Baesens (2012), “A multi dimensional quality assesment of state-of-the-art process discovery algorithms using real-life event logs”, *Journal of Information Systems*, 37 (7) (2012), 654–676
- [9] Ishak H.A. Meddah, Khaled Belkadi, Mohamed Amine Boudia (2016), “Parallel mining small patterns from business process traces”, *International Journal of Software Science and Computational Intelligence*, 8 (7) (2016), 654–676
- [10] Ishak H.A. Meddah, Khaled Belkadi (2016), “Discovering patterns using process mining”, *International Journal of Rough Sets and Data Analysis*, 3 (4) (2016), 21–31
- [11] Apache Spark (2019.05.14) , “A unified analytics engine for large-scale data processing”, Apache Software Foundation, <http://spark.apache.org/>, Access Date: 2019.05.14
- [12] Lars Frank (1999), “Integrity Problems in Distributed Accounting Systems with Semantic ACID Properties”, *Third International Conference on Integrity and Internal Control in Information Systems*, Springer Science and Business Media LLC, 147–159
- [13] S. M. Kawale, A. N. Holambe, J.D. Bokefode (2017), “A review on big data concepts and various analytic techniques”, *International Journal of Computer Trends and Tecnology*, 52 (1) (2017), 13–16
- [14] Business Oriented Architecture (2019.05.15) , “BOA Banking Platform used in 20 countries”, <https://www.kuveytturk.com.tr/en/about-us/about-kuveyt-turk/news/kuveyt-turk-speeds-up-its-investments-in-technology-and-transformation-through-architect>, Access Date: 2019.05.15
- [15] Kuveyt Turk API Banking Developer Portal (2019.05.15) , “Kuveyt Turk API Market”, <https://developer.kuveytturk.com.tr/#/> , Access Date: 2019.05.15
- [16] Charu C. Aggarwa (2017), “Outlier Analysis”, *Springer International Publishing AG*
- [17] Kuveyt Turk Public GIT Repository (2019.05.15), “Outlier Detection and Analysis of Variance with Apache Spark SQL”, <https://github.com/KuveytTurk/ProcessMining/blob/master/src/main/java/com/kuveytturk/pm/com/kuveytturk/pm/anova/AnovaUtility.java>, Access Date: 2019.05.15
- [18] Fredrick Gravetter and Larry B. Wallnau (2009), “Statistics for the Behavioral Sciences”, *Wadsworth Cengage Learning Publishing*
- [19] Kuveyt Turk Public GIT Repository (2019.05.15), “Linear Regression Analysis with Apache Spark SQL”, <https://github.com/KuveytTurk/ProcessMining/blob/master/src/main/java/com/kuveytturk/pm/com/kuveytturk/pm/regression/LinearRegression.java>, Access Date: 2019.05.15
- [20] Apache Hadoop (2019.05.14) , “Apache Hadoop: Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models”, Apache Software Foundation, <https://hadoop.apache.org/>, Access Date: 2019.05.14
- [21] Hadoop MapReduce (2019.05.14) , “Hadoop MapReduce: A software framework for easily writing applications which process vast amounts of data (multi-terabyte data-sets) in-parallel on large clusters (thousands of nodes) of commodity hardware in a reliable, fault-tolerant manner”, Apache Software Foundation, <https://hadoop.apache.org/docs/stable/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html>, Access Date: 2019.05.14
- [22] Apache Sqoop (2019.05.14) , “Apache Sqoop: A tool designed for efficiently transferring bulk data between Apache Hadoop and structured datastores such as relational databases.”, Apache Software Foundation, <https://sqoop.apache.org/>, Access Date: 2019.05.14
- [23] Apache Hive (2019.05.14) , “The Apache Hive: Data warehouse software facilitates reading, writing, and managing large datasets residing in distributed storage using SQL”, Apache Software Foundation, <http://hive.apache.org/>, Access Date: 2019.05.14
- [24] Hadoop Distributed File System (HDFS) (2019.05.14) , “HDFS: A distributed file system designed to run on commodity hardware”, Apache Software Foundation, <http://hadoop.apache.org/hdfs/>, Access Date: 2019.05.14